

Quick Check

Lesson 7: Factors Affecting Population Size: Why Are the Fish Disappearing?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Ecology | Lesson 7

Learners sorted factors that could reduce fish numbers in a Kenyan river into 'living' and 'non-living'. Which choice correctly gives one biotic factor and one abiotic factor?

- A** Biotic: water temperature; abiotic: disease and parasites
- B** Biotic: drought lowering water level; abiotic: algae cover
- C** Biotic: dissolved oxygen; abiotic: competition among tilapia
- D** Biotic: Nile perch preying on small fish; abiotic: low dissolved oxygen in the water

After reading about population growth and limiting factors, which statement best describes the carrying capacity of a river for tilapia?

- A** The maximum number of tilapia that the river's resources can support over time
- B** The number of tilapia that fishermen are allowed to catch each season
- C** The number of tilapia born in the river each year
- D** The number of different fish species living in the river

In a simulation, a group first runs a baseline with fish and food only. In trial 2 they add predators and change nothing else. The fish population ends lower than in the baseline. Why can they conclude that predators caused the drop?

- A** Because the graph in trial 2 has more lines than the graph in the baseline
- B** Because predators are always the most important factor in any ecosystem
- C** Because only one variable was changed, so the predators are the only difference between the trials
- D** Because the fish population was lower at the start of trial 2

In the PhET simulation, learners keep the fish but reduce the amount of food available. What do they most likely observe on the fish population graph?

- A** The fish population stops growing and falls to a lower level, because food is a limiting factor
- B** The fish population rises quickly, then the food increases
- C** The fish population keeps rising because there are no predators
- D** The fish population stays exactly the same as the baseline

A river receives fertiliser and sewage from farms and homes upstream. Fish numbers fall while algae and weeds increase and the water smells bad. Which explanation best links these observations?

- A** Excess nutrients cause an algae boom. When the algae die and decompose, dissolved oxygen falls and fish cannot survive.
- B** Algae cover lets more light reach the bottom of the river, which makes the fish leave.
- C** Algae are poisonous, so they kill the fish directly as soon as they appear.
- D** Extra nutrients give the fish more food, so the fish grow old and die in large numbers.

Tilapia in a dam breed every few months, yet their population stops increasing at about 5 000 fish and stays near that number. What is the best explanation?

- A** The tilapia choose to stop breeding when the dam looks full
- B** Limiting factors such as food, oxygen and space prevent the population from growing beyond what the dam can support
- C** Population size can only grow until it reaches 5 000 in any water body
- D** Predators are the only factor that can control a population

Learners are adding human activities to their river ecosystem model and linking them to nutrient input. Which human activity should they connect to extra nutrients in the water?

- A** Nile perch preying on small fish
- B** Drought lowering the water level
- C** Tilapia competing for food
- D** Fertiliser from farms and untreated sewage washing into the river

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).